UNIVERSITY OF GRAZ
Department of Mathematics
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MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/2026

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EXERCISE SHEET 1

Due: 19. Mar 2025, 10:00

EXERCISE 1.1
We consider the following chemical reaction with kinetic constant $\lambda > 0$,

$$C_4H_8 \; + \; C_2H_6O \quad \xrightarrow{\lambda} \quad C_6H_{14}O$$

where the reaction between isobutylene (C_4H_8) and ethanol (C_2H_6O) give ethyl tert-butyl ether $(C_6H_{14}O)$. Denote a, b, c the concentrations of (C_4H_8), (C_2H_6O) and $(C_6H_{14}O)$, respectively.

1. Write the systems of ODEs that describes the chemical reaction above. Then, prove that

$$\dot{c} = \lambda(a_0 - c)(b_0 - c), \qquad \text{with } \; a(0) = a_0, \; b(0) = b_0. \tag{1}$$

2. Perform the qualitative analysis of (1) (steady states, stability analysis, convexity, asymptotic behavior). Consider the cases $a_0 > b_0$, $a_0 < b_0$ and $a_0 = b_0$.

3. Determine the solution of (1) and confirm the results obtained in 2.

EXERCISE 1.2
Consider the following ODE

$$\dot{x} = \lambda x(1 - x) - k, \qquad \lambda, k > 0. \tag{2}$$

a. Interpret the model (2). Then, investigate the stability analysis of (2) with respect to the parameters λ, k.

b. Draw the phase space and interpret the result.

EXERCISE 1.3
Consider the following ODE

$$\dot{x} = x - 2\mu x^2 + x^3, \qquad \mu \in \mathbb{R}. \tag{3}$$

1. Discuss the existence of the steady states with respect to the value of μ and identify the critical value μ_c.

2. Determine the stability of the steady states when $\mu > \mu_c$.

3. Verify the assumptions such that (3) admits a tangent bifurcation.

EXERCISE 1.4 (Optional)
Perform the heuristic derivation to prove that the chemical reaction

$$A \quad + \quad B \quad \longrightarrow \quad C,$$

with rate constant $\lambda > 0$, is described by the following system of ODEs

$$\begin{cases} \dot{a} = -\lambda ab, \\ \dot{b} = -\lambda ab, \\ \dot{c} = \lambda ab. \end{cases}$$

Hint: bring inspiration from the heuristic derivation performed during the class for the linear case. More specifically, assume that particles of type A and particles of type B are (uniformly) distributed among N cells. Then introduce appropriate assumptions to determine the probability that the two different types of particles interact with each other within a single cell.